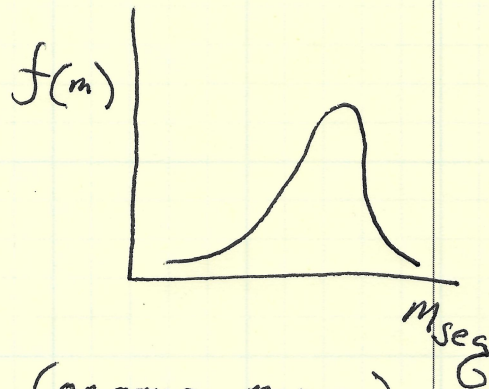


EMPIRICAL MAXIMUM LIKELIHOOD

⇒ APPLY ML TECHNIQUE TO NONPARAMETRIC DATA

e.g. CENTIPEDE SEGMENTS
(INTEGER RV)

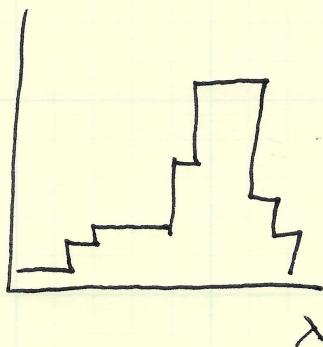


Q CAN I DETERMINE THE MEAN (OR OTHER MOMENTS)
OF AN RV DISTRIBUTION FOR WHICH I DON'T
HAVE AN A PRIORI PARAMETRIZATION?

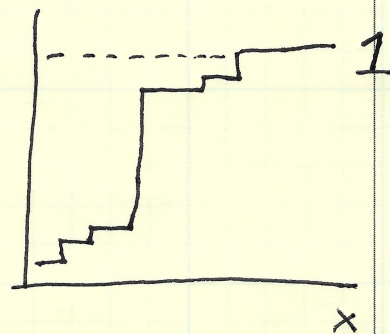
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WHAT DO I KNOW FOR NONPARAMETRIC DATA?

EPDF
(HISTOGRAM,
DIVIDE BY
 $N \Delta x$)



ECDF



p_i = fraction of all events
w/in bin i

$$L = \prod_{i=1}^N p_i$$

$$\sum_{i=1}^N p_i = 1$$

WHAT IS L_{MAX} ?



$\Rightarrow$ Consider $a + b = \delta = \text{constant}$

$\hookrightarrow$ WHAT IS $\max a \cdot b$?

$$\frac{\partial}{\partial a} (a \cdot b) = \frac{\partial}{\partial a} (a(\delta - a)) = 0$$

$$\delta - 2a = 0$$

$$a = b = \delta/2$$

- $a \cdot b$ is maximum when $a = b$
- generalizes to ANY number of terms
- IN ECDF PLOT, THIS CORRESPONDS TO
ALL STEPS BEING EQUAL $\rightarrow$ UNIFORM DISTRIBUTION!

$$\begin{aligned} L &= \sum_{i=1}^N p_i \\ \sum_{i=1}^N p_i &= 1 \\ L_{\max} &= \sum_{i=1}^N \frac{1}{N} \end{aligned}$$

WHAT ABOUT NON-UNIFORM DISTRIBUTIONS?

⇒ Suppose I'm told that $\mu_0 = \langle x \rangle$ is mean of data.
 This is a CONSTRAINT on my system of equations.

Namely,

$$\langle x \rangle = \sum_{i=1}^N p_i x_i = \mu_0$$

$$\sum_{i=1}^N p_i (x_i - \mu_0) = 0$$

NOTE: $g(x, \mu_0) = x - \mu_0$

"ESTIMATING FUNCTION"

DEPENDS ON MOMENT

I WISH TO DETERMINE

We need to MAXIMIZE L AND get p_i ($= \tilde{p}_i$)

SUBJECT TO A CONSTRAINT

$$\frac{L}{L_{\max}} = \frac{\prod \tilde{p}_i}{\prod \frac{1}{N}} \leftarrow \text{OUR DISTRIBUTION}$$

$$= \prod N \tilde{p}_i \leftarrow \text{UNIFORM DISTRIBUTION}$$

(LATER, we'll see ADVANTAGES TO THIS APPROACH)

CONSTRAINED MAXIMIZATION $\longleftrightarrow$ LAGRANGE MULTIPLIERS

Rewrite $\tilde{p}_i = \frac{1}{[1 + \lambda(x_i - \mu_0)]^N}$ $\lambda \equiv \text{LAGRANGE MULTIPLIER}$

$\neq$ CONSTANT
Equation becomes $\sum_{i=1}^N \frac{(x_i - \mu_0)}{[1 + \lambda(x_i - \mu_0)]^N} = 0$

(Solve to find λ)

Ignore, constant

$$\frac{L}{L_{\max}} = \prod_i \tilde{p}_i = \prod_i \frac{1}{[1 + \lambda(x_i - \mu_0)]^N}$$

$$\ln\left(\frac{L}{L_{\max}}\right) = - \sum_i \ln(1 + \lambda(x_i - \mu_0))^N$$

"It has been shown" (Owen, 1980s) that

$$W = -2 \ln\left(\frac{L}{L_{\max}}\right)$$

Converges to a χ^2 distribution in $N_{df} = \# \text{ constraints}$ (7)

→ Our goal is to minimize W

Evaluate W for each possible value of μ

Find $W(\mu)$ at which 95% cutoff occurs!

< NOTEBOOK me - example, i pynb >